

Working with PI Controllers under Linux Operating Systems

This document gives a summary of notes regarding the usage of Physik Instrumente software on Linux operating systems. Some software features still have to be adapted to Linux operating systems. The list below shall be understood as temporarily since we permanently improve the Linux software. Please write to info@pi.de to find out the current availability of the features.

For detailed information on how to communicate with PI controllers from ARM-based platforms see *A000T0089-PI Software on ARM-Based Platforms*, which can be downloaded from our website at www.pi.ws.

Legal Information

Software products provided by PI may incorporate and/or make use of third-party software components. For more information, please refer to the Third Party Software Note linked below.
[General Software License Agreement](#)
[Third Party Software Note](#)

General Limitations

- **GPIO Interface:**
The GPIO (General Purpose Interface Bus) interface can only be used with our GCS LabVIEW driver set. The GCS library does not yet support the GPIO interface.
- **Write permissions:**
If you want to modify files which are not located in your home directory, you first have to change the write permission for these files (e.g. for LabVIEW vi files).

Limitations for LabVIEW

- **Linux Distributions** supported by National Instruments:
For the Linux distributions supported by National Instruments see their corresponding [web page](#).
- **Analog GCS LabVIEW driver set:**
These drivers have not yet been tested and are therefore not supported.

Technical Note

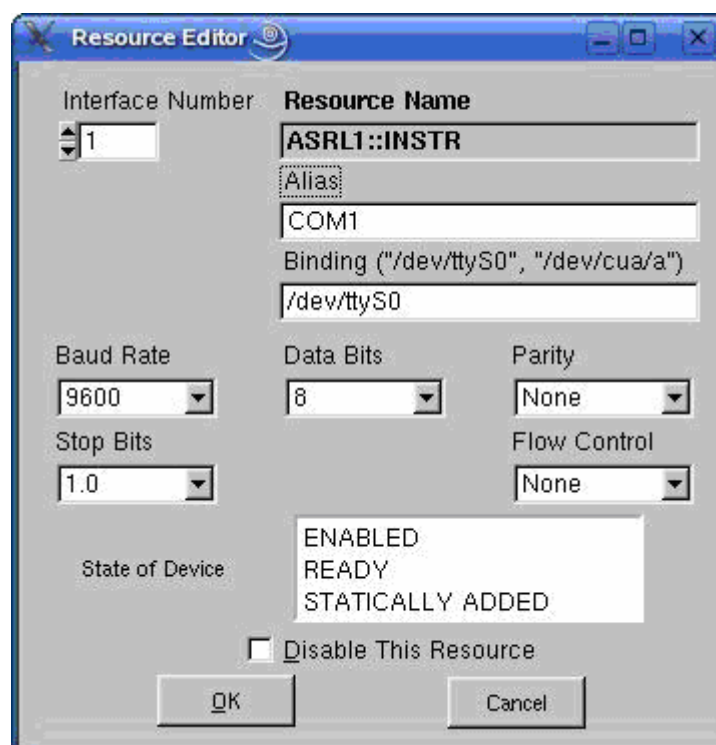
A000T0021, valid for LINUX PCs

MRa, BRo, ASt, 8/8/2023

- **libPIGraph3D:**
The libwxPIGraph3D.so shared object is not yet able to run on all of the most common Linux distributions. If you want to use vi files which make use of libwxPIGraph3D.so (eg. 2D Scan.vi and Show_Save_Load_XYZ_Data.vi), please look for software updates on our website or contact our customer service.
- **Using Serial Ports in LabVIEW under Linux:**
When you are working with LabVIEW on an Linux PC, you have to enable the serial ports before you can use them. To do this, use the visaconf tool which is installed with LabVIEW. By default, it is located in
`/usr/local/vxipnp/linux/NIvisa/visaconf`

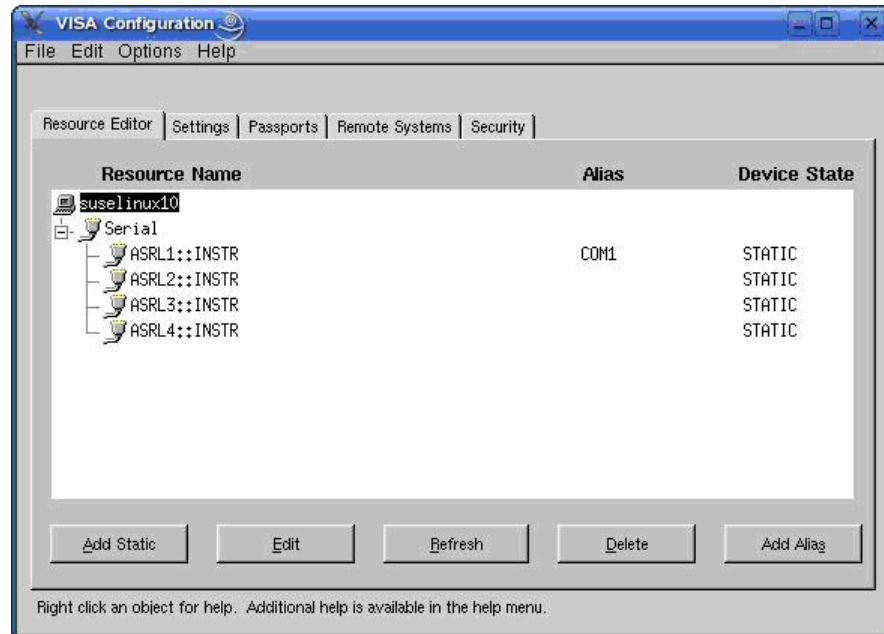
Proceed as follows:

1. Open the visaconf tool.
2. In the "VISA Configuration" window, open the "Resource Editor" tab. Select the serial port you want to configure, and click the **Edit** button. The "Resource Editor" window opens.
3. In the "Resource Editor" window, enter COMn in the "Alias" field, where n is the number of the COM port (n is arbitrary, but there has to be a COMn entry in the list of interfaces in the LabVIEW connection dialog). We recommend you use COM1.



Leave all other settings unchanged and confirm the selection with **OK**.

4. In the "VISA Configuration" window, check the assignment of ASRLm::INSTR to COMn on the "Resource Editor" tab.



5. Close the visaconf tool and, when prompted to confirm your changes, click **Yes**.

